

BLE Medical Device Hardware-Layer Architecture Reference

Cross-brand systems analysis for unified microservice design · v1.0 · 2026-07-15

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1. Executive Summary

This repository contains protocol and product documentation for building a **unified hardware-layer microservice** that connects to multiple proprietary smart medical devices and behaves like their official companion apps.

1.1 Brands / folders analyzed

Folder	Brand / ecosystem	What is actually documented
beurer/	Beurer GmbH	Full BLE transmission protocol for BM54 blood pressure monitor (rev01–rev03)
FORA/	ForaCare / FORA	Product brochure only for FORA 6 Connect — no BLE protocol
nipro/SH3/	Nipro SH3 hub + multi-vendor sensors	Mixed pack: A&D UA-651BLE , Masimo MightySat , TaiDoc-style thermometer TICD , SIG HTP , Nipro Cocoron ECG , SH3 platform feature list

1.2 Architectural finding (one sentence)

Medical BLE peripherals in this corpus fall into **two session archetypes**:

1. **Episodic dump devices** (BP meters, many thermometers): advertise when they have data → short secure session → push/pull store → disconnect.
2. **Configured stream devices** (SpO2, ECG): connect → host configures → continuous or periodic notify stream → long-lived or supervised session.

A single naive “connect and subscribe” path **will fail** on A&D (5-second arming gate), thermometer (dual-command wake), MightySat (framed proprietary protocol), and Cocoron (always-on HR heartbeat).

1.3 Implementability from this corpus alone

Device	Protocol completeness	Can implement companion-like driver?
Beurer BM54	High	Yes
A&D UA-651BLE	High	Yes
Masimo MightySat	High	Yes
NT-100B Thermometer (TICD)	High (app-layer serial)	Yes (BLE transport + serial framing)
SIG HTP	High (profile reference)	Yes as generic thermometer collector
Nipro Cocoron ECG	Medium (behavior/design; UUID/payload IF external)	Partial — need interface workbook or sniffer
FORA 6 Connect	None (brochure)	No — blocked

2. Document Corpus & Coverage

2.1 File inventory

Beurer (beurer/)

File	Role
BM54_transmissionprotocol_rev01_20190805.pdf	Initial BM54 BLE protocol
BM54_transmissionprotocol_rev02_20200430.pdf	BPM field changes
BM54_transmissionprotocol_rev03_20210716.pdf	Authoritative latest — advertising data update

FORA (FORA/)

File	Role
FORA 6 Connect Brochure-Ver1.0_20250407.pdf	Marketing brochure (Bluetooth mentioned; no GATT)

Nipro SH3 (nipro/SH3/)

File	Role
■■■■■■■■■■sdk_ble_ua-651ble_ v1.2_141027_-En.pdf	A&D UA-651BLE Application Development Spec v1.2
■SpO2■MightySat Customer Support Document and Interface Protocol - CSD-1322B.pdf	Masimo MightySat BLE protocol CSD-1322 Rev B
■■■■■■■■■■1-1■■■■■■■■① Communication Protocol T1CD_Thermometer_v1.16_20210216.pdf	NT-100B thermometer serial/BLE command ICD v1.16
■■■■■■■■■■1-2■■■■■■■■② BLUETOOTH PROFILE SPECIFICATION 9.■■■■■■■■■■HTP_V10.pdf	Bluetooth SIG Health Thermometer Profile v1.0
Cocoron■■■■■■■■■■.pdf (and duplicate)	Nipro ECG (Cocoron) body software design Rev 2.0
SH3_Features.pdf / SH3_FeatureCompact.pdf	SH3 hub/app platform features (not radio protocol)
■■■■■■■■■■SaMD_DTx...pptx	SaMD/DTx trends — out of BLE scope

2.2 What each document type contributes

Doc type	Contribution to microservice design
Vendor BLE protocol PDFs	UUIDs, pairing, sequences, parse layouts
SIG HTP	Generic thermometer collector behavior, idle timeouts, reconnection
Cocoron software design	Always-on model, HR period, ECG cadence, battery/LOD policy
SH3 features	Outer hub lifecycle (scan → assign → MQTT), not per-device GATT
FORA brochure	Product capability only (6 analytes, 1000 memory, Bluetooth)

3. Cross-Brand Similarity & Difference Matrix

3.1 Master comparison (Core Extraction Targets)

Dimension	Beurer BM54	FORA 6 Connect	A&D UA-651BLE	Masimo MightySat	Thermometer T1CD NT-100B	Cocoron ECG	SIG HTP (ref)
Product role	Upper-arm BP	Multi-parameter BG meter	Upper-arm BP	Finger SpO2 multi-param	IR non-contact thermometer	Wearable ECG + HR	Generic thermometer profile
GAP role	Peripheral / GATT server	Unknown	Peripheral / GATT server	Peripheral / GATT server	Peripheral (BLE v4) / Classic SPP (v2/v3)	Peripheral	Peripheral
Autonomous silent fetch	Yes — auto Indicate all stored BP after connect	Unknown	Gated auto Indicate after CCCD+time within 5 s	No — host must configure streaming / request trends	No — host poll after dual-wake	Yes for HR/ECG after config; continuous model	Yes — Indicate when measurement ready

Historical sync	Bulk Indicate every stored record (both users)	1000 records claimed	Bulk Indicate oldest-first; optional custom memory cmds	Get Trend Record by session ID (0–100)	Index poll 0x25/0x26 after count 0x2B	Not classic store-download; streaming dominant	Multiple Temperature Measurement indications
Real-time	Not primary (ICP Notify unused)	Unknown	Not primary path in SDK	Params 1 Hz + waveforms ~31.25 Hz after Configure Streaming	Host can start IR measure 0x41 (some models)	HR periodic + ECG 10 s / hourly / continuous abnormal	Intermediate Temperature Notify optional
Wake / fetch command	None — connect only	Unknown	CCCD=0x0002 + DateTime write (mandatory gates)	Framed cmds on RX char (0x01/0x02/0x03/0x04/0x05/0x06/0x07)	Two arbitrary cmds in 10 s; optional 0x54 notify	App config (HR period, ECG on) after worn	CCCD enable for Indicate
Keep-alive	Episodic; preferred CI 100–200 ms	Unknown	5 s / 30 s idle timeouts; device disconnects after dump	Streaming traffic is keep-alive	Short session; no ping	HR notify is heartbeat; silence → device reset	Idle terminate ~5 s
Pairing model	6-digit Passkey (newer)	Unknown	SM Security Request + encryption + bond; Unpair 0x10	Not documented	Classic PIN 111111 / SSP or 0000; BLE pairing not fully specified	Multi-phone blocked; security in higher-level reqs	Bonding recommended; re-encrypt each connect
Profile model	SIG BLP 0x1810 + DIS	Unknown	SIG BLP + Battery + DIS + proprietary 0xF000	Proprietary 128-bit service + framed messages	Proprietary serial GATT 0x1523/0x1524	Custom Renesas profile (DateTime, ECG Config)	SIG Health Thermometer Service + DIS
Invalid / quality	Measurement status bits	Unknown	SFLOAT error sentinels; status bits; no-time flag	System + parameter exception bitmasks; trend 255	Checksum / ACK mismatch; limited clinical flags	LOD; battery <2.3 V; HR clamp 0–350	RFU handling rules
Low battery	Not in BM54 protocol	“1000 times” marketing	Battery Service 0x180F / 0x2A19	Not a dedicated battery GATT in CSD	Not specified in T1CD	BLE notify + LED patterns	Not defined by HTP itself
Mid-transfer drop	Reconnect + re-Indicate remaining (practical)	Unknown	Store residual; send next successful session	Ordinal gap detect; re-arm stream; resume trend IDs	Re-poll by index	Reconnect policy; packet loss tolerated by IF design	Link-loss re-advertise / re-scan

3.2 Similarity clusters (what can be shared)

Cluster	Members	Shared pattern
SIG Blood Pressure Indicate dump	Beurer BM54, A&D UA-651BLE	Service 0x1810, char 0x2A35 Indicate, flags + SFLOAT + optional timestamp/pulse/user/status
Episodic measurement peripherals	Beurer, A&D, HTP thermometers	Advertise when ready → short connect → transfer → disconnect
Framed proprietary command channel	MightySat, T1CD thermometer	Host writes requests; device responds on notify/write channel with checksummed frames
Configured clinical stream	MightySat, Cocoron	Host sets what/how often; device pushes asynchronously
Device Information Service	Beurer, A&D, HTP (and typical SIG devices)	Manufacturer, model, serial, FW/HW revision strings
Time synchronization gate	A&D (hard), MightySat SetClock, T1CD 0x33	Historical timestamps require host clock write

Manufacturer Specific advertising	Beurer CID 0x0611, Masimo CID 0x0243	Prefer Company ID filter over optional service UUID
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3.3 Difference clusters (what must diverge)

Difference	Why it matters
Passkey (Beurer) vs SM bond (A&D) vs undocumented (MightySat) vs Classic PIN (thermometer legacy)	Pairing module cannot be one-size-fits-all
Auto-dump vs 5 s gated dump vs poll-index vs configure-stream	Session arming is driver-specific
SIG health profiles vs full proprietary GATT	Discovery and parsers branch early
Episodic BP vs always-on ECG	Connection manager policies opposite
Quality via BP status bits vs Masimo exception bitmasks vs LOD voltage	Normalization table required
FORA unknown	Cannot join any cluster until protocol obtained

3.4 Service / characteristic presence matrix

Service / capability	Beurer	FORA	A&D	MightySat	TICD thermo	Cocoron	HTP
GAP 0x1800	Yes	?	Yes	(implied)	(implied)	Yes	Yes
GATT 0x1801	Yes	?	Yes	?	?	Yes	Yes
Blood Pressure 0x1810	Yes	?	Yes	No	No	No	No
Health Thermometer (HTS)	No	?	No	No	Optional dual path	No	Yes (profile)
Battery 0x180F	No (not documented)	?	Yes	No (in CSD)	No	Custom battery notify	Optional
Device Information 0x180A	Yes	?	Yes	Via cmd 0x01	Via cmds 0x24/0x27/0x28	Version features	Yes
Pulse Oximeter 0x1822	No	?	No	Optional in adv only	No	No	No
Proprietary custom service	No	?	Yes 0xF000	Yes 128-bit	Yes 0x1523	Yes custom	No
Date/Time characteristic	No dedicated (timestamp in measurement)	?	Yes 0x2A08	SetClock cmd	Clock cmds 0x23/0x33	Date Time char	Optional in measurement

3.5 Session archetype matrix

Device	Archetype	Typical session length	Who terminates
Beurer BM54	Episodic dump	Seconds—tens of seconds	Not fully specified; expect short session after dump
A&D UA-651BLE	Episodic dump (gated)	Seconds; hard 5 s / 30 s timers	Device after memory send complete
MightySat	Configured stream + on-demand history	Minutes while streaming	Host or link loss
TICD thermometer	Host-pollled short session	Seconds per poll sequence	Host (or power-off 0x50)
Cocoron ECG	Always-on stream	Hours/days while worn	Link loss / reset / user

HTP thermometer	Episodic dump	~5 s idle window	Either side after idle
FORA	Unknown	Unknown	Unknown

4. Brand / Device Deep Dives

4.1 Beurer BM54

4.1.1 Identity

Field	Value
Manufacturer	Beurer GmbH
Model	BM54 blood pressure monitor
Transport	Bluetooth Low Energy
Role	GAP Peripheral / GATT Server
Advertising name	BM54
Company ID (Mfg Specific Data)	0x0611 (Beurer)
Passkey feature flag in adv	0x01 in Mfg payload; newest also documents 0x11060103
Primary protocol doc	BM54_transmissionprotocol_rev03_20210716.pdf

4.1.2 Services and characteristics

Primary services

Spec name	Assigned number	Full UUID
Generic Access	0x1800	00001800-0000-1000-8000-00805f9b34fb
Generic Attribute	0x1801	00001801-0000-1000-8000-00805f9b34fb
Blood Pressure	0x1810	00001810-0000-1000-8000-00805f9b34fb
Device Information	0x180A	0000180A-0000-1000-8000-00805f9b34fb

Generic Access

Characteristic	UUID	Properties	Notes
Device Name	0x2A00	Read	"BM54"
Appearance	0x2A01	Read	0 Unknown
Peripheral Privacy Flag	0x2A02	Read, Write	0x00
Reconnection Address	0x2A03	Write	—
Peripheral Preferred Connection Parameters	0x2A04	Read	CI 100–200 ms; Slave Latency 0; Supervision Timeout Multiplier 1000

Generic Attribute

Characteristic	UUID	Properties
Service Changed	0x2A05	Indicate

Blood Pressure service

Characteristic	UUID	Properties	Notes
Blood Pressure Measurement	0x2A35	Indicate	Primary data path
Intermediate Cuff Pressure	0x2A36	Notify	Not used by BM54
Blood Pressure Feature	0x2A49	Read	Always [0x04, 0x00] — irregular pulse supported

Device Information (selected)

Characteristic	UUID	Notes
Manufacturer Name String	0x2A29	Read
Model Number String	0x2A24	Read
Serial Number String	0x2A25	Read
Hardware / Firmware / Software Revision	0x2A27 / 0x2A26 / 0x2A28	Read
System ID	0x2A23	All zeros documented
IEEE 11073-20601 Regulatory Cert	0x2A2A	Includes ASCII "Experimental" style payload in doc
PnP ID	0x2A50	TI company 0x000D; Product Id 0; Product Version 272

4.1.3 Advertising

- Name: BM54
- Manufacturer Specific Data type 0xFF
- Payload pattern: first two bytes company ID 0x0611 (little-endian presentation as 0x1106... in doc tables), then passkey-feature identifier 0x01
- Newest generation documents extended form including trailing 0x03 (0x11060103)

4.1.4 Pairing & security

- Newer BM54 versions use 6-digit Passkey Entry for connection.
- Older behavior not fully differentiated beyond advertising passkey flag.
- Multiple bond feature: not supported in BP Feature bitfield (bit clear).

4.1.5 Autonomous data fetching

Documented behavior:

The BM54 is sending the data automatically after a connection has been established successfully. This can be triggered by clicking the M1 or the M2 Button on the device. It is also triggered after every measurement. The device is sending all data from both users. For every measurement stored on the BM54 an indication is sent.

Implications for companion-like behavior:

1. No proprietary "wake" write is required.
2. Host must enable Indications on 0x2A35 (CCCD bit for Indicate).
3. Expect N indications for N stored measurements.

4. Both user memories are sent in one session.
5. Intermediate cuff pressure is not part of the product path.

4.1.6 Historical vs real-time

Mode	Support
Historical store dump	Yes — all stored measurements via Indicate
Real-time cuff inflation stream	No (0x2A36 unused)
Live continuous BP	No

4.1.7 Keep-alive

- Preferred connection interval: **100.00 ms – 200 ms**
- Slave latency: **0**
- Supervision timeout multiplier: **1000**
- No ping/dummy-read documented.
- Design assumes **short dump session**, not long keep-alive.

4.1.8 Revision deltas (must track in parser)

Rev	Date	Change
01	2019-08-05	Initial
02	2020-04-30	Blood Pressure Measurement changes
03	2021-07-06	Advertising data changes
User ID mapping	rev01 vs rev03	rev01 text used user values 1 and 2; rev03 uses 0x00 = user1, 0x01 = user2

4.2 FORA 6 Connect

4.2.1 Identity (product only)

Field	Value from brochure
Model	FORA 6 Connect
Parameters	Blood Glucose, Hematocrit, Hemoglobin, β -Ketone, Uric Acid, Total Cholesterol
Connectivity	Bluetooth
App	iFORA HM (free download)
Memory	1000 measurement results with date/time
Power	1× 1.5V AAA; battery life “1000 times”
Dimensions / weight	89.8 × 54.9 × 18 mm; 46.1 g without batteries
BG range	10–600 mg/dL (0.55–33.3 mmol/L)
Meal tags	AC, PC, AutoQC
Strip series	BG-HCT-HB, KB, UA, TCH (sample volumes and reaction times listed)

4.2.2 BLE protocol status

Required extraction target	Status in folder
GATT services/characteristics	Missing
Pairing model	Missing
Autonomous fetch mechanism	Missing
Historical sync sequence	Missing
Keep-alive	Missing
Error/battery signaling	Missing
Data parse layouts	Missing

4.2.3 Microservice implication

Driver cannot be implemented from current documentation.

Place FORA in a **blocked adapter** slot until ForaCare protocol/SDK or BLE sniffer capture vs iFORA is available. Do **not** assume TaiDoc-family framing (even if industry-adjacent) without evidence.

4.3 A&D UA-651BLE (Nipro SH3 pack)

4.3.1 Identity

Field	Value
Manufacturer	A&D Company, LTD.
Model	UA-651BLE blood pressure monitor
Doc	Application Development Specification for A&D Bluetooth BLE Series v1.2 (2014-10-27)
Stack model	SIG profiles + A&D proprietary profile

4.3.2 Services

Service	UUID	Purpose
GAP	0x1800	Standard
GATT	0x1801	Standard
Blood Pressure	0x1810	Measurement Indicate + Feature + Date Time
Device Information	0x180A	Identity strings + System ID + Regulatory
Battery	0x180F	Battery Level
A&D Custom	0xF000	Settings/commands (20-byte R/W characteristic)

Documented custom characteristic UUID fragment (128-bit):

233BF001-5A34-1B6D-975C-000D5690ABE4 style mapping in profile table (base family with 233BF000... service context).

4.3.3 Blood Pressure Measurement (0x2A35)

Uses standard BLP structure with A&D fixed/flagged fields:

- Pulse Rate Flag: **present (fixed)**
- User ID Flag: **not present (fixed)** → User ID unknown/255 semantics when absent

- Measurement Status Flag: **present (fixed)**
- Measurement Status is **Little Endian**
- Body Movement Detection Flag fixed to “no body movement” in feature/status constraints as documented
- Irregular Pulse supported (Feature value example 0x0C, 0x00)

Error encodings (critical):

Condition	Encoding
Measurement Error	C1/C2/C4 value = 0xFF, 0x07 (SFLOAT NaN-style error)
Pulse Rate Range Error	C4 = 0x00, 0x08; LCD shows “E”; upper/lower detection possible
Not Time Settings	Flags bit1 = 0; timestamp N/A
Irregular Pulse example status	C6 = 0x04, 0x00

4.3.4 Blood Pressure Feature (0x2A49)

Little Endian. Documented support emphasizes irregular pulse; multiple bonds **not supported (fixed)**. Example irregular pulse support value 0x0C, 0x00.

4.3.5 CCCD

- Blood Pressure Measurement is sent with **Indication**
- CCCD Indicate enable value: 0x02, 0x00 (little endian)

4.3.6 Date and Time (0x2A08)

- Used to set/read device clock
- Year/month/day/hour/minute/second fields with range checks
- **If any field out of range, device does not update clock**
- A&D requires time sync **at every connection and pairing**
- After battery change without time adjust, measurements may be stored/sent **without** date/time
- UA-651BLE **does not support reading time** via 0x2A08 nor custom cmd 0x04 (document restriction)

4.3.7 Battery Service

Level examples	Value
100%	0x64
66%	0x42
40%	—
33%	0x21

Devices can take a measurement when they can still send battery information (device-defined thresholds).

4.3.8 Device Information (selected)

Field	Example / rule
Manufacturer	A&D Medical (even number of chars required for strings)

Model	UA-651BLE
Serial	e.g. 5140700001
System ID	BD address with FFFE insertion pattern
Regulatory	Continua-style IEEE 11073 certification blob

4.3.9 Pairing & communication sequence (official)

Pairing mode:

1. App enters pairing scan mode.
2. User puts device into pairing per manual (switch on / hold ~3 s as diagrammed).
3. Advertise limited discoverable (~60 s window in diagram).
4. Connect; detect services: BP 0x1810, Custom 0xF000, DIS 0x180A, Battery 0x180F, Date Time 0x2A08.
5. Security Request → Pairing Confirm → Encryption.
6. Read DIS/Battery/DateTime as needed.
7. Write settings via custom service (time, etc.).
8. After successful pairing device shows “End”.
9. Idle timeout if no commands: **30 seconds** → disconnect.

After measurement (data transfer mode):

1. Measurement finishes → undirected connectable advertising (~60 s).
2. App should scan with aggressive interval to catch advertising quickly.
3. Connect + security/encrypt.
4. Within **5 seconds after encrypted** (or after needed write request):
 - Write Date Time
 - Write CCCD = **0x0002** (Indications enabled)
5. Once **both** CCCD and Date Time are done, device goes to **Send Memory Data immediately** (without waiting remaining timeout).
6. Device Indicates stored measurements **oldest first**.
7. Host confirms each Indication (ATT).
8. During Send Memory Data, device **does not accept settings**.
9. After finished, host may set CCCD back to 0x0000; device **disconnects after send complete**.
10. If CCCD/time not received in time: data **remains stored** for next successful connection.
11. After encrypted, if no command: **5 s** timeout disconnect.

4.3.10 Custom service command model

20-byte readable/writable buffer:

Field	Meaning
Size	Length from Type to end of Value
Type	0 = Read cmd, 1 = Write cmd, 2 = Response (peripheral)

Command	Command ID
Value	Payload

Documented UA-651BLE commands (partial table recoverable from PDF):

ID	Operation
0x01	Set Time
0x03	Request disconnection by device / disconnect to stand-by
0x10	Unpair and stand-by
0x12	Delete all memory
0xA6	Number of memory / buffer related
0xD6	Read buffer size (example request/response in doc)
0xE1	Request all memory data (shown in sequence diagrams)

Set Time value format example fields: year lower 2 digits, month, day, hour, minute, second.

4.3.11 Preferred connection parameters (profile mapping)

From attribute table notes:

- Peripheral Preferred Connection Parameters example values in mapping: 0x0050, 0x00A0, 0x0000, 0x0258 style (interval/latency/timeout fields as mapped in A&D profile table)

4.4 Masimo MightySat (Nipro SH3 pack)

4.4.1 Identity

Field	Value
Manufacturer	Masimo Corporation
Product	MightySat
Doc	CSD-1322 Revision B / CO-067126
Transport	BLE asynchronous proprietary protocol
Endianness	Little Endian
Company ID	0x0243

4.4.2 Advertising (critical discovery rule)

Do not rely on Service UUID filtering — PLXP service UUID presence varies.

Filter using **Manufacturer Specific Data**:

Field	Value
Company ID	0x0243
Structure Version	0x01
Product Type	0x01
Product Variant	0x00 Consumer (SpO2, PR, PI); 0x01 RX (adds PVi, RRp)

Serial Number	uint32
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Two adv formats:

1. **With PLXP support:** includes Service UUID 0x1822 + name MightySat + Mfg data
2. **Without PLXP:** name + Mfg data only (no Service UUID field)

4.4.3 Proprietary GATT service

Item	Value
128-bit Service UUID	54c21000-a720-4b4f-11e4-9fe20002a5d5
RX characteristic (host → device)	16-bit ID 0x1001, Write Without Response
TX characteristic (device → host)	16-bit ID 0x1002, Notify only

4.4.4 Message framing

SOM LEN PAYLOAD[N] CRC 0x77 N d1..dN crc8	
Field	Rule
SOM	Always 0x77
LEN	Size of (PAYLOAD + CRC); does not include LEN itself; valid 1–255
CRC	CRC-8-CCITT over bytes after LEN and before CRC
CRC poly	$P(x) = x^8 + x^2 + x + 1$
CRC seed	0
Multi-byte ints	LSB first

4.4.5 Commands

Command	ID	Direction	Purpose
Get Device Information	0x01	Host→Dev	SW version, available params/waveforms, trend meta, session IDs
Set Clock	0x02	Host→Dev	Unix epoch seconds (uint32)
Configure Streaming Data	0x03	Host→Dev	Enable param/waveform bitmasks
Get Trend Record	0x06	Host→Dev	One trend session by ID
Clear Trend Records	0x07	Host→Dev	Wipe all trends

ACK/NACK

Type	ID	Meaning
ACK	0xFE	Success; data byte = original command ID
NACK	0xFF	Failure; command ID + error code

Error codes

Code	Name
0	None

1	Record Not Found
2	Command Failed
3	Not Supported
4	Record Corrupted
5	Bad CRC

4.4.6 Get Device Information response (payload concept)

- Command ID 0x01
- SW Version (2)
- Available Parameters bitmask (2)
- Available Waveforms bitmask (1)
- Trend version (2)
- Trended Parameters bitmask (2)
- Number of trend records uint8 **[0–100]**
- Oldest Trend Session Id (4)
- Current Trend Session Id (4)

Parameter bits: 0 SpO2, 1 PR, 2 PVI, 3 PI, 4 RRp

Waveform bits: 0 Pleth, 1 SIQ

Example command frame: [0x77, 0x02, 0x01, 0x07]

4.4.7 Configure Streaming → real-time path

- Payload: Parameters bitmask (2) + Waveforms bitmask (1)
- Example: [0x77, 0x05, 0x03, 0x1F, 0x00, 0x03, 0xD6]
- On ACK: device starts periodic transfer

Parameter Streaming response (async, ~1 Hz)

- Response ID **0x05**
- System Exceptions bitmask (4 bytes)
- For each enabled parameter (LSB→MSB order): exception byte + value
- Values uint8 except **PI = uint16 ×100**

System exception bits (documented):

Bit	Meaning
21	Sensor Off Patient
22	Pulse Search
23	Interference Detected
24	Low Perfusion

Parameter exception bits:

Bit	Meaning
0	Low Confidence
2	Invalid
4	Startup State

Waveform Streaming response (async)

- Response ID **0x04**
- Ordinal uint8 (wraps; detects missing packets)
- Samples: Pleth int8 + SIQ
- Sampling period **32 ms** (~31.25 Hz)
- SIQ bit7 = Invalid; bits0–6 = 0–100 quality-ish value

4.4.8 Historical trends

Rules:

- Do **not** request next record before previous response/NACK.
- Trend Session ID increments when user removes device and values freeze.
- **PI is not saved** as trend even if streamed.
- Invalid sample value = **255**.
- Each param trend record: Min, Max, Average, Last (4× uint8).
- Response includes session id, duration seconds, timestamp of last sample (Unix epoch).

4.4.9 Pairing / keep-alive

- Pairing/bonding **not specified** in CSD-1322B.
- No dedicated ping command.
- Keep-alive = configured streaming notifications.
- Integrity = frame CRC + ordinal continuity.

4.5 Non-Contact Thermometer NT-100B / TICD (Nipro SH3 pack)

4.5.1 Identity

Field	Value
Document	Meter Interface Control Document, Thermometer Meter NT-100B, Ver 1.16 (2021-02-16 / history to 2018-04-30)
Roles	GW = gateway/hub/computer; MD = medical device
Physical serial baseline	RS232-like framing over link

4.5.2 Transport variants

Bluetooth generation	Connection model	Pairing notes
v2 meter	SPP	PIN 111111 ; always stands by for pairing/connection
v3 meter	SPP	Must pair first; SSP preferred; if no SSP use PIN 0000
v4 meter (BLE)	GATT	UUID base 1212-efde-1523-785feabcd123; Service 0x1523 ; Characteristic 0x1524 (write/notify)

4.5.3 Entering communication mode (wake)

Critical companion-app behavior:

1. Establish connection channel.
2. Send **first two adjoining commands within 10 seconds** to ensure device is in communication mode.
3. Commands may be arbitrary (example 0x24).
4. **Skip responses** whose ACK command does not match the request.
5. Some models send unsolicited **0x54 Notification for entering communication mode** — host must handle at start and not confuse with request responses.
6. TD1241 does not support 0x54 / 0x41 / 0x52 (model exceptions).

4.5.4 Frame structure

Byte: 1 2 3..n-2 n-1 n Name: Start CMD Index/Addr/Data Stop CheckSum GW→MD: 0x51 CMD ... 0xA3 sum(1..n-1)
GW←MD: 0x51 ACK ... 0xA5 sum(1..n-1)

- Checksum: 8-bit sum of bytes 1..n-1
- Typical command frames: 8 bytes

Serial parameters (RS232 baseline in doc): baud 19200, 8N1.

4.5.5 Full command list

CMD	Name	Direction	Notes
0x23	Read device clock time	Both	To minute packing tables
0x24	Read device model	Both	Project code
0x25	Read storage data part 1 (time)	Both	Index 0 = latest
0x26	Read storage data part 2 (result)	Both	Object + ambient temps
0x27	Read serial number part 1	Both	SN_0..SN_3
0x28	Read serial number part 2	Both	SN_4..SN_7
0x2B	Read storage number of data	Both	Count word
0x33	Write system clock time	Both	Same packing as 0x23
0x41	Start IR temperature measurement	Both	Not on TD1241
0x50	Turn off device	Both	ACK field notes 0x55 in one table row
0x52	Clear/delete all memory	Both	Not on TD1241

0x54	Entering communication mode notify	MD→GW only	Unsolicited
------	---------------------------------------	------------	-------------

4.5.6 Storage read semantics

- Index 0x0000 = latest reading.
- Part1 (0x25): date, minute, hour, temperature category type, unit (°C/°F).
- Types (3-bit category): ear, forehead, rectal, armpit, object surface, room, children.
- Part2 (0x26): object temperature and ambient/background temperature in **0.1°C units** (when unit Celsius path).
- Historical sync is **fully host-polled**, not device push.

4.5.7 Real-time

- Command **0x41** starts an IR measurement and returns object + ambient temperatures.
- Not supported on all model variants.

4.6 Bluetooth SIG Health Thermometer Profile (HTP) Reference

(Included in Nipro folder as profile baseline for non-contact thermometer integration.)

4.6.1 Roles

Role	Stack role
Thermometer	GATT Server, GAP Peripheral
Collector	GATT Client, GAP Central

Services: **one** Health Thermometer Service + Device Information Service.

4.6.2 Collector mandatory capabilities

- Discover HTS
- Discover Temperature Measurement + its CCCD
- Configure **Indications**
- Receive **multiple** indications (stored measurements)
- Support with/without Time Stamp and Temperature Type fields
- Support °C and °F
- Ignore RFU flag bits and unknown trailing octets

Optional: Intermediate Temperature notifications, Measurement Interval R/W, Temperature Type read.

4.6.3 Connection establishment patterns

Informative typical scenario:

Thermometer stays powered off between uses; advertises when it has data; collector scans (often whitelist); thermometer sends indications/notifications; thermometer terminates when done.

Advertising recommendations

Phase	Adv interval
First 30 s (fast)	20–30 ms
After 30 s (reduced power)	1–2.5 s

Collector scan recommendations

Phase	Scan interval / window
First 30 s	30–60 ms interval, 30 ms window
Later option 1	1.28 s / 11.25 ms
Later option 2	2.56 s / 11.25 ms

Fast connection interval for discovery/encrypt: 50–70 ms min/max, then switch to preferred params.

Idle connection: either side may terminate if idle **> 5 seconds**.

Bonded reconnect: collector shall start encryption after each connection; if encryption fails, re-bond, rediscover if needed, re-write CCCD.

Link loss: thermometer re-enters connectable advertising; collector restarts connection procedures.

4.6.4 Security

- Thermometer/collector security considerations defined in HTP §6 (bonding recommended for repeated use).
- Use as policy baseline for thermometer-class episodic devices when vendor is silent.

4.7 Nipro Cocoron ECG

4.7.1 Identity

Field	Value
Document	NIPROA01-SD-002 Rev 2.0 (2019-03-12) — body software design
Product domain	ECG development / Cocoron communication
MCU	Renesas RL78/G1D
BLE stack	Renesas RBLE + RWKE RTOS
Phone role	iPhone Central; device Peripheral
Power	3V coin cell

4.7.2 Functional capabilities

#	Function	Behavior
1	ECG waveform send	125 Hz sample → Huffman compress → BLE send
2	Heart rate notify	Period selected by iPhone: 5 / 10 / 30 / 60 s ; 60 s moving average RR; HR range 0–350 (clamp)
3	Battery monitor	Sample every 1 min ; notify phone; LED patterns

4	LOD (leads-off / not worn)	Check every 1 s ; notify on change; ECG/HR start only after return-to-worn
5	BLE connection	app peripheral middleware; BD address + serial from code flash; block 2nd simultaneous iPhone with warning

4.7.3 ECG transmission policy (companion-like)

Condition	ECG behavior
App launch / connect baseline	Send 10 seconds of ECG
Normal ongoing	10 seconds every 1 hour
HR outside app-configured normal range and abnormal ECG-send enabled	Continuous ECG until battery depletes

4.7.4 Always-on model & keep-alive

- Device is designed for **persistent BLE communication** while worn.
- Regular HR notifications act as **application-level heartbeat**.
- If HR notifications stop for a defined period: **device self-reset** (life-safety oriented recovery).
- Communication errors recover primarily by **BLE reconnect**, not local logging.

4.7.5 Battery & LED

Condition	Threshold / pattern
Battery OK	$\geq 2.3 \text{ V}$
Battery abnormal	$< 2.3 \text{ V}$
LED normal	1 flash / 5 s (ON 10 ms)
LED abnormal	3 flashes / 5 s (ON 10 ms, OFF 200 ms)

4.7.6 Error handling policy

Category	Example	Recovery
Device-alone	Deadlock	WDT self-reset
Device-alone	BLE packet send error	Wait/discard; stop continuous stream if prolonged; re-enable GAP
Link	Out of range	GAP enable/disable around communications; seamless reconnect
Link	Packet loss	Format designed to limit blast radius (no TCP retransmit)
Link	No communication	Silence detection → reset

4.7.7 Characteristics mentioned (incomplete UUID map)

Design sequences reference:

- Date Time Characteristic
- ECG Config Characteristic

Exact UUIDs and on-wire payload layouts are not fully present in this PDF (referenced external interface design workbook / xlsx). Behavior and timing are specified.

4.7.8 Non-worn recovery

- On return from non-worn: clear previous HR average; after **5 seconds** of new averaging, send HR.

4.8 SH3 Hub Platform Context

SH3_Features.pdf is **not** a radio protocol, but defines the product envelope your microservice sits in:

Hub operator capability	Relevance
BLE scan / discover sensors	Outer discovery loop
Register sensors with backend	Inventory
Assign sensors to patients	Multi-user routing (critical for Beurer dual-user dumps)
Auto collect BLE health data	Driver orchestration
MQTT publish (hubId, sensorId, patientId, vitals, timestamp)	Northbound bus
Heartbeat every 30 s	Hub health (not device ping)
Supported sensor types	HR, SpO2, Temperature, BP (matches this datasheet pack)

Use SH3 as **deployment topology**; use sections 4.1–4.7 as **device drivers**.

5. End-to-End Workflows (Discovery → Disconnection)

This section is the companion-app behavioral specification for the microservice.

5.1 Global state machine (unified)

[Idle/Scan] → identify candidate (name / CID / service UUID) → [Connecting] → [Pairing/Bonding/Encrypting] (policy by driver) → [Service Discovery] → [Session Arming] (CCCD / time / dual-wake / configure stream) → [Historical Sync] and/or [Realtime Streaming] → [Optional Control] (clear memory, set clock, power off) → [Teardown] (disable CCCD / stop stream / disconnect) → [Idle/Scan] or [Bonded Wait] On error: [Recover] → reconnect / re-arm / resume cursor

5.2 Discovery workflow (all brands)

5.2.1 Common discovery steps

- Start LE scan (and Classic inquiry only if thermometer legacy SPP required).
- Parse advertising/scan response:
 - Local Name
 - Service UUIDs
 - Manufacturer Specific Data (Company ID + payload)
- Score against device registry.
- Apply whitelist for bonded peripherals when re-connecting episodic devices.
- Prefer **fast scan** first 30 s when expecting a post-measurement advertisement (HTP guidance; A&D “catch advertising ASAP”).

5.2.2 Per-brand discovery filters

Device	Primary filter	Secondary	Anti-pattern
Beurer BM54	Name BM54 + CID 0x0611	Passkey flag in Mfg data	Ignoring passkey-capable flag
A&D UA-651BLE	Service 0x1810 + custom 0xF000 after connect	Name/model via DIS	Assuming any 0x1810 is A&D
MightySat	CID 0x0243 + name MightySat	Variant byte; serial	Filtering only on 0x1822
TICD thermo BLE	Service 0x1523 / base UUID	Name from manual	Treating as HTP-only
HTP thermo	HTS UUID in adv (should)	Local name	Forcing bond before first connect always
Cocoron	Name/custom (from missing IF doc)	BD address + serial in flash	Treating as episodic BP
FORA	Unknown	App-paired name unknown	Guessing TaiDoc UUIDs

5.3 Pairing / bonding workflow

5.3.1 Beurer BM54

Scan BM54 (note passkey flag) → Connect → If newer FW: Passkey Entry (6 digits shown on device/UI) → Bond keys stored by host OS / microservice secret store → Proceed to service discovery

5.3.2 A&D UA-651BLE

User puts device in pairing mode (manual) → Limited discoverable advertise (~60 s) → Connect → Security Request / Pairing / Encryption → Write time + any production/settings via custom service → Device displays "End" on success → Disconnect / stand-by Later measurements use bonded undirected connectable advertising Unpair command 0x10 available to clear bond side on device

5.3.3 MightySat

Connect to advertised MightySat → (Pairing policy not documented – implement defensively: try OS default LE security; capture official app if fails) → Enable Notify on TX 0x1002 → Optional GetDeviceInfo / SetClock

5.3.4 Thermometer TICD

If Classic v2: pair PIN 111111, SPP connect If Classic v3: pair (SSP or 0000), SPP connect If BLE v4: GATT connect to 0x1523/0x1524 → Dual arbitrary commands within 10 s → Handle optional 0x54 → Continue command session

5.3.5 HTP collector

Prefer bond on first configuration → On each reconnect: Start encryption to verify bond → If encrypt fails: user re-bond, rediscover if needed, rewrite CCCD

5.3.6 Cocoron

Connect single phone only → If second central attempts: send warning info → Configure DateTime / ECG Config characteristics → Ensure worn (LOD) → Start HR/ECG paths

5.4 Service discovery & capability read

Step	Shared action	Brand specifics
1	Discover primary services	SIG vs proprietary branches
2	Discover characteristics + CCCDs	Mandatory for Indicate/Notify paths

3	Read DIS	Beurer/A&D/HTP; MightySat uses cmd 0x01 instead/in addition
4	Read feature bits	BP Feature 0x2A49; MightySat param bitmasks
5	Read battery if present	A&D 0x2A19; Cocoron battery notify

5.5 Session arming workflow (critical)

This is where companion apps “silently” unlock data.

5.5.1 Beurer arming

Write CCCD of 0x2A35 = Indications enabled (0x0002) → Wait for N Indications (all stored records, both users) → Confirm each Indication

5.5.2 A&D arming (hard real-time gate)

After Encrypted: t0 = now Write Date Time (0x2A08 and/or custom 0x01) Write CCCD 0x2A35 = 0x0002 Must complete required gates within 5 seconds of encrypt → Device enters Send Memory Data (oldest first) If timeout: no send; data retained

5.5.3 MightySat arming

Enable Notify TX → GetDeviceInfo (0x01) → SetClock (0x02) if historical timestamps needed → For realtime: ConfigureStreaming (0x03) with bitmasks → For history: iterate GetTrendRecord (0x06) from oldest..current session IDs (never pipeline next before response)

5.5.4 TICD thermometer arming

Send cmd A (e.g. 0x24) Send cmd B within 10 s total window Ignore mismatched ACKs Optionally observe 0x54 → Write clock 0x33 if needed → Read count 0x2B → For i in 0..count-1: 0x25(i), 0x26(i)

5.5.5 HTP arming

Write CCCD Temperature Measurement = Indications enabled Optional: enable Intermediate Temperature notifications Optional: set Measurement Interval non-zero for periodic indications → Receive one or more indications

5.5.6 Cocoron arming

Confirm LOD = worn → Set HR notification period (5/10/30/60) → Enable ECG path per policy (10s now / hourly / continuous-on-abnormal) → Receive HR + compressed ECG packets

5.6 Historical sync workflows

Beurer

Armed Indicate session → For each Indication: parse BPM record → emit Reading → Deduplicate by (timestamp, userId, systolic, diastolic, pulse) if re-sent → Map userId 0/1 to patient binding → End when indications stop / idle timeout / disconnect

A&D

Armed gated session → Receive Indications oldest-first with ATT Confirm → Parse including error sentinels → Optionally query memory count via custom cmds before/after → Do not clear memory until durable store commit → Optional clear 0x12 only after success → Expect device disconnect when done

MightySat

```
info = GetDeviceInfo for sessionId in [oldest .. current] (or known missing set): resp =
GetTrendRecord(sessionId) if NACK Record Not Found: continue/skip parse min/max/avg/last per available params if
value == 255: mark INVALID
```

TICD

```
count = ReadStorageNumber (0x2B) for index in 0 .. count-1: t = ReadStoragePart1 (0x25, index) v =
ReadStoragePart2 (0x26, index) join into single Reading (time + object/ambient + type + unit)
```

HTP

While indications arrive: parse Temperature Measurement flags → fields emit Reading Optional Measurement Interval for future periodic indications

Cocoron

No full “download all ECG history” in design doc Optional: capture streaming windows as time-series artifacts Persist HR time series continuously while connected

FORA

BLOCKED – unknown

5.7 Real-time streaming workflows

MightySat realtime

ConfigureStreaming(paramsMask, waveMask) loop: on Notify: demux by response ID 0x05 → parameter sample @1Hz + exceptions 0x04 → waveform chunk + ordinal gap check if ordinal discontinuity: flag packet loss; optional resync policy on link loss: reconnect → re-enable notify → re-configure streaming

Cocoron realtime

Receive HR at configured period (heartbeat) Receive ECG bursts per policy If HR silence beyond threshold: expect device reset; reconnect If out-of-range HR + ECG-on: expect continuous ECG flood – size buffers accordingly

TICD “realtime measure”

Send 0x41 Wait matching ACK with temps (Not continuous stream)

Beurer / A&D realtime cuff

Not supported as product path (Beurer ICP unused; A&D path is post-measure dump)

5.8 Control operations workflows

Operation	Beurer	A&D	MightySat	TICD	Cocoron
Set clock	Not as separate char (timestamps come from device)	Required each session	Cmd 0x02	Cmd 0x33	Date Time characteristic
Clear memory	Not documented	Custom 0x12	Cmd 0x07	Cmd 0x52	N/A (stream device)
Power off	Not documented	Disconnect/stand-by cmds	Not documented	Cmd 0x50	Always-on design
Unpair	Host-side unbond	Cmd 0x10	Host-side	Host-side	Host-side

Request all memory	Automatic on connect	Auto after arm; also 0xE1 in diagrams	Trend iterator	Index iterator	N/A
--------------------	----------------------	---------------------------------------	----------------	----------------	-----

5.9 Disconnection & teardown workflows

Episodic devices (Beurer, A&D, HTP)

If dump complete: optional disable CCCD accept device-initiated disconnect (A&D explicit) return to bonded scan/whitelist wait for next measurement advertise If dump incomplete: mark resume needed reconnect when device re-advertises re-arm session rely on residual storage (A&D/Beurer store until successful transfer)

Streaming devices (MightySat, Cocoron)

StopStreaming / disable notifications (if applicable) Clear host stream state Disconnect gracefully On unexpected drop: exponential backoff reconnect; restore config

TICD

Optional Clear memory after commit Optional Turn off 0x50 Close GATT/SPP

5.10 Link-loss recovery matrix

Device	Device-side recovery	Host-side recovery
Beurer	Re-advertise on next user action / measurement (inferred)	Rescan; reconnect; re-enable Indicate
A&D	Retain memory; advertise after measure again	Fast scan; re-encrypt; re-time; re-CCCD in 5 s
MightySat	Continues locally; trends retained until cleared	Re-notify; GetDeviceInfo; re-stream; resume trend IDs
TICD	Stays with stored readings	Reconnect; dual-wake; resume index
HTP	Re-enter connectable adv with recommended intervals	Reconnect procedures; re-encrypt if bonded
Cocoron	GAP enable cycles; WDT; silence reset	Auto reconnect; re-apply ECG/HR config; check LOD

6. Data Parsing Specifications (Per Brand)

6.1 Common normalized Reading model (logical)

Field	Description
deviceId	Stable identity (serial / BD_ADDR / System ID)
brand	beurer / aand / masimo / ticd / cocoron / fora / htp
metric	bp_systolic, bp_diastolic, pulse, spo2, pr, pi, temp_object, hr, ecg_samples, ...
value	Numeric SI-ish value after conversion
unit	mmHg, bpm, %, °C, mV, ...
timestamp	UTC if convertible; else device-local + quality flag
userId	Beurer multi-user; others often single
quality	Normalized flags (see §8)

raw	Original bytes + parse path id for audit
-----	--

6.2 Beurer BM54 — Blood Pressure Measurement Indicate payload

Flags byte always 0x1E for BM54 meaning present: timestamp, pulse, user ID, measurement status; unit mmHg path.

Field	Size	Format	Notes
Flags	1	bitfield	always 0x1E
Systolic	2	SFLOAT	1 == 1.0 mmHg; range 0–300
Diastolic	2	SFLOAT	same
MAP	2	SFLOAT	always 0 / not supported
Timestamp	7	year uint16, month, day, hours, minutes, seconds	year 1582–9999
Pulse	2	SFLOAT	0–255
User ID	1	uint8	rev03: 0x00 user1, 0x01 user2
Measurement Status	2	bitfield	see below

Example (from doc):

[0x1E 0x70 0x00 0x4D 0x00 0x00 0x00 0xDF 0x07 0x01 0x0E 0x0A 0x37 0x00 0x48 0x00 0x01 0x00 0x00]
→ SYS 117, DIA 77, MAP 0, 2015-01-14 10:55:00, pulse 72, user2 (rev03 mapping), status no error.

SFLOAT decode note: Bluetooth MEDFLOAT16 — parser must implement IEEE-11073 SFLOAT, not raw int16.

Measurement status bits:

Bit field	Meaning
body movement	0 none / 1 movement
cuff fit	0 proper / 1 too loose
irregular pulse	0 none / 1 detected
pulse rate range	00 in range / 01 high / 10 low / 11 RFU
measurement position	0 proper / 1 improper

Feature characteristic: [0x04, 0x00] → irregular pulse feature supported only (among the set).

6.3 A&D UA-651BLE — BPM + custom

6.3.1 BPM Indicate

Same general BLP layout as SIG with A&D constraints:

- Pulse present fixed
- User ID not present fixed
- Status present fixed; little endian
- Parse error sentinels before numeric conversion:

Sentinel	Meaning
C1/C2/C4 = 0xFF 0x07	Measurement error
C4 = 0x00 0x08	Pulse range error

6.3.2 Custom 20-byte command buffer parse

```
[Size][Type][Command][Value...] Type: 0 read, 1 write, 2 response
```

Parse responses by Command ID; examples in doc for time set and buffer size (0x06).

6.4 Masimo MightySat — framed messages

6.4.1 Ingress demux

```
assert SOM == 0x77 len = LEN payload = next (len-1) bytes crc = last byte verify CRC-8-CCITT(payload) # over
bytes after LEN before CRC per spec wording cmd = payload[0] switch cmd: 0x01 device info response 0x04 waveform
stream 0x05 parameter stream 0x06 trend record 0xFE ACK 0xFF NACK
```

6.4.2 Parameter stream value order

Enabled bits from **LSB to MSB** define field order in the packet.

PI special-case: uint16, divide by 100.0 for display value.

6.4.3 Trend record

Per parameter block: min, max, avg, last (uint8).

Map 255 → INVALID.

Attach sessionId, duration, lastSampleUnix.

6.5 TICD thermometer — 8-byte command frames

6.5.1 Validate frame

```
start == 0x51 stop == 0xA3 (host) or 0xA5 (device) checksum == sum(bytes[0..n-2]) & 0xFF
```

6.5.2 Clock packing (0x23/0x33)

- Data_1||Data_0: Year(7) | Month(4) | Day(5)
- Data_2/Data_3: minute (6) / hour (5) with reserved bits per tables

6.5.3 Storage part1 (0x25)

- Index little-endian request
- Response includes D_M_Year, minute+type, hour+type+unit packing (Table C)
- Temperature category enum 0..6
- OutUnit 0=°C, 1=°F

6.5.4 Storage part2 (0x26)

- Object temperature uint16, unit 0.1°C (when Celsius path)
- Ambient/background uint16, unit 0.1°C

6.5.5 Serial number

- 16 hex characters: SN from 0x28 (high) + 0x27 (low) as documented example order

6.6 HTP Temperature Measurement

Follow Bluetooth HTS characteristic:

- Flags determine unit, timestamp presence, temperature type presence
- Temperature value is FLOAT (IEEE-11073 32-bit), not SFLOAT
- Collector must accept missing optional fields

6.7 Cocoron ECG / HR

Stream	Parse notes
HR	Scalar bpm 0–350; period-driven; average from 60 s RR window
ECG	125 Hz samples Huffman-compressed; requires external IF tables for bit packing
Battery	Voltage-oriented notify; threshold 2.3 V
LOD	Boolean worn/not; gate other streams

Without the external interface workbook, ECG sample reassembly cannot be fully specified here. HR/battery/LOD behaviors can still drive session logic.

6.8 FORA

No parse specification available.

6.9 Multi-user / patient routing rules

Device	Rule
Beurer	Every dump includes both users — filter by User ID before MQTT publish to patient topic
A&D	Single-user style (User ID flag fixed absent)
Others	Single patient assignment at hub

7. Keep-Alive, Timeouts & Link Management

7.1 Timeout catalog

Timer	Device	Value	Effect
Preferred CI	Beurer	100–200 ms	Connection param target
Supervision multiplier	Beurer	1000	Link supervision
Pairing idle	A&D	30 s	Disconnect if no commands after encrypt in pairing
Post-measure arming	A&D	5 s	Must CCCD+time or no dump
Post-confirm idle	A&D	5 s	Disconnect paths in sequence diagrams
Advertising windows	A&D diagrams	~60 s	Pairing / post-measure advertise
Dual-command wake	TICD	10 s	Enter communication mode

HTP idle	HTP	5 s	Either side may disconnect
HTP fast adv	HTP	30 s then slower	Discovery power tradeoff
HTP fast CI	HTP	50–70 ms	During discovery/encrypt
MightySat param rate	MightySat	1 Hz	Stream keep-alive
MightySat wave period	MightySat	32 ms	Waveform pacing
Cocoron HR period	Cocoron	5/10/30/60 s	App-level heartbeat
Cocoron battery poll	Cocoron	1 min	Battery notify cadence
Cocoron LOD poll	Cocoron	1 s	Wear state
Cocoron ECG baseline	Cocoron	10 s / hour	Burst policy
SH3 hub MQTT heartbeat	Hub	30 s	Hub liveness (not device)

7.2 Keep-alive strategies to implement

Strategy name	When to use	Mechanism
NoneEpisodic	Beurer, A&D, HTP, TICD	Complete transfer; allow disconnect
TrafficStream	MightySat	Maintain ConfigureStreaming; watch 1 Hz
AppHeartbeatHR	Cocoron	Expect HR notify within 2–3× period; else recover
HubHeartbeat	SH3	MQTT 30 s — orthogonal to BLE

No device in this corpus documents a dummy GATT read ping. Do not invent one unless sniffer shows the official app doing so.

7.3 Connection interval policy

1. Use **fast intervals** during service discovery, bonding, encryption, and bulk indicate dumps.
2. After setup, accept peripheral preferred parameters (Beurer 100–200 ms; A&D mapped prefs; HTP guidance).
3. For high-rate waveform (MightySat ~31.25 Hz samples in chunks), ensure interval/throughput supports Notify payload rates.
4. For Cocoron continuous ECG abnormal mode, budget for sustained throughput + Huffman frames.

8. Error, Quality & Battery Signaling

8.1 Normalized quality enum (recommended)

OK BODY_MOVEMENT CUFF_LOOSE IRREGULAR_PULSE PULSE_OUT_OF_RANGE IMPROPER_POSITION NO_TIMESTAMP MEASUREMENT_ERROR
SENSOR_OFF PULSE_SEARCH INTERFERENCE LOW_PERFUSION LOW_CONFIDENCE INVALID STARTUP LEADS_OFF LOW_BATTERY CRC_FAIL
NACK_* PARTIAL_TRANSFER

8.2 Mapping tables

Beurer status → normalized

Source bit	Normalized
body movement	BODY_MOVEMENT
cuff too loose	CUFF_LOOSE

irregular pulse	IRREGULAR_PULSE
pulse high/low	PULSE_OUT_OF_RANGE
improper position	IMPROPER_POSITION
all clear	OK

A&D → normalized

Source	Normalized
0x07FF measurement fields	MEASUREMENT_ERROR
pulse 0x0008 encoding	PULSE_OUT_OF_RANGE
timestamp flag 0	NO_TIMESTAMP
irregular pulse status	IRREGULAR_PULSE
battery low discrete levels	LOW_BATTERY (policy threshold)

MightySat → normalized

Source	Normalized
bit21 Sensor Off	SENSOR_OFF
bit22 Pulse Search	PULSE_SEARCH
bit23 Interference	INTERFERENCE
bit24 Low Perfusion	LOW_PERFUSION
param Low Confidence	LOW_CONFIDENCE
param Invalid / value 255	INVALID
param Startup	STARTUP
NACK codes	NACK_*
Bad CRC	CRC_FAIL

TICD → normalized

Source	Normalized
checksum fail	CRC_FAIL
mismatched ACK	PARTIAL_TRANSFER / protocol error
(clinical invalid not richly coded)	OK unless transport error

Cocoron → normalized

Source	Normalized
LOD not worn	LEADS_OFF
V < 2.3	LOW_BATTERY
HR silence recovery	PARTIAL_TRANSFER / link recovery event
HR > 350 clamped	note as clamped value

8.3 Mid-transfer failure playbooks

Brand	Playbook
Beurer	On disconnect, keep last fully confirmed indication only; on reconnect accept full dump; dedupe
A&D	Trust device storage; next successful 5 s arming resends residual; never clear early
MightySat	Use ordinal gaps to drop incomplete wave segments; re-request trends by session id
TICD	Resume index loop from last successful index
Cocoron	Reconnect + reconfig; do not assume gapless ECG across reset
HTP	Standard link-loss reconnect; re-enable CCCD if bond lost

9. Unified Microservice Abstraction

9.1 Logical architecture layers



9.2 Driver interface (logical)

```
DeviceDriver { id, brand, archetype matchAdvertisement(adv) -> score pairingPolicy() -> PairingMode
discover(conn) armSession(conn, options) // CCCD/time/wake/configure syncHistory(conn) -> iterator[Reading]
startRealtime(conn, mask) -> subscription stopRealtime(conn) readDeviceInfo(conn) readBattery(conn) -> optional
clearMemory(conn) // dangerous; policy gated setTime(conn, ts) onLinkLoss(conn) teardown(conn) }
```

9.3 Orchestrator algorithm (companion-app equivalent)

```
loop forever: adv = scan.next() driver = registry.bestMatch(adv) if driver is null: continue conn = connect(adv)
try: driver.pairIfNeeded(conn) driver.discover(conn) driver.setTime(conn) if driver.requiresTimeGate
driver.armSession(conn) if policy.wantsHistory: for reading in driver.syncHistory(conn): normalize -> route to
patient -> publish durableAck() maybeClearMemory(driver) // only after durableAck if policy.wantsRealtime and
driver.supportsRealtime: driver.startRealtime(conn) supervise(heartbeats/timeouts) else: driver.teardown(conn)
catch LinkLoss: driver.onLinkLoss(conn) scheduleReconnect(driver, bondedAddress)
```

9.4 Archetype engines

EpisodicEngine (Beurer, A&D, HTP, TICD poll)

- Aggressive scan when expecting post-measure advertising
- Short connection
- Strict arming deadlines (A&D 5 s, TICD 10 s)
- History first; no long-lived keep-alive
- Device or host disconnect after completion

StreamEngine (MightySat, Cocoron)

- Maintain connection while clinically needed

- Configuration transaction at start
- Watchdog on expected notify cadence
- Backpressure buffers for ECG/waveform floods
- Reconfigure after reconnect

9.5 Configuration surface (per installed device)

Config key	Purpose
pairingMode	Passkey / SM / PIN / none
armTimeoutMs	Driver-specific
historyMode	push-indicate / poll-index / trend-session
realtimeMask	Params/waveforms/HR period
autoClearMemory	default false
patientBinding	including Beurer userId map
qualityPolicy	which flags reject vs accept-with-warning

10. Reusable Modules Inventory

This is the reuse plan for the microservice codebase (module boundaries — not code).

10.1 Highly reusable (use across many brands)

Module	Responsibility	Reused by
BleScanner	LE scan, whitelist, duty-cycle fast/slow profiles (HTP tables)	All BLE devices
AdvertisementParser	Name, service UUIDs, Mfg data TLV	All
CompanyIdRegistry	Map CID → brand (0x0611 Beurer, 0x0243 Masimo, ...)	Discovery
GattClientFacade	Connect, MTU, discover, read/write, CCCD, indicate/notify callbacks	All GATT devices
SecurityManagerFacade	Just Works / Passkey / bond storage / re-encrypt on connect	Beurer, A&D, HTP, others
CccdHelper	Encode 0x0001 Notify / 0x0002 Indicate; enable/disable	Beurer, A&D, HTP, streams
SfloatCodec	IEEE-11073 SFLOAT encode/decode	Beurer, A&D BP
Float11073Codec	32-bit medical FLOAT	HTP temperature
BlpMeasurementParser	Flags-driven BP measurement parse skeleton	Beurer + A&D (with hooks)
DisReader	Standard Device Information characteristics	Beurer, A&D, HTP
BatteryServiceReader	0x180F / 0x2A19	A&D (+ future SIG devices)
SessionOrchestrator	Archetype engines + retries	All
ReadingNormalizer	Metric/unit/quality canonicalization	All
PatientRouter	Map device+userId → patientId	Beurer multi-user + hub assign
DurableInbox	At-least-once store before memory clear	A&D/MightySat/TICD clears

LinkWatchdog	Expected notify cadence supervision	MightySat, Cocoron
ReconnectPolicy	Backoff, bonded direct connect, scan fallback	All
TimeSyncService	Write device clocks in vendor formats	A&D, MightySat, TICD, Cocoron
MqttPublisher	SH3 northbound schema	Hub integration
AuditRawFrameLog	Raw bytes for regulatory/debug	All (esp. medical)

10.2 Partially reusable (shared with adapters)

Module	Shared core	Brand-specific adapter
BloodPressureDriverBase	BLP parse + Indicate dump loop	Beurer auto-arm vs A&D 5 s gate + custom service
EpisodicDumpEngine	advertise→connect→dump→disconnect	HTP / Beurer / A&D timing params
FramedCommandChannel	request/response correlation, timeout	MightySat CRC framing vs TICD checksum framing
TrendHistoryCursor	iterate IDs with ack gating	MightySat sessions; TICD indices
QualityMapper	enum + bit extractors	per-brand bit dictionaries
PairingWizard	UX/state for PIN/passkey	different PIN sources

10.3 Brand-specific modules (do not force-share)

Module	Why isolated
BeurerPasskeyAdvertisingDecoder	Unique Mfg flag generations
BeurerUserIdRevMapper	rev01 vs rev03 user id semantics
AndCustomServiceCodec	20-byte Size/Type/Cmd buffer + cmd set
AndFiveSecondArmingGate	Hard real-time correctness
MightySatCrc8Ccitt	Specific poly/seed and LEN definition
MightySatStreamDemux	0x04/0x05 async interleaving
TicdDualWake	10 s two-command + 0x54 ignore rules
TicdBitPackedClockTables	Non-obvious date bit packing
CocoronHuffmanEcgDecoder	Unique compression; external tables
CocoronLodGate	Wear gating before streams
ForaDriver	Placeholder blocked

10.4 Reusable test fixtures / golden vectors

Fixture	Source
Beurer example indication bytes	BM54 protocol example
MightySat example command frames	CSD examples ([0x77, 0x02, 0x01, 0x07], etc.)
A&D error SFLOAT sentinels	SDK error section
TICD frame checksum examples	Construct from sum rule
HTP idle/scan timing tables	HTP §5

10.5 Suggested package map


```
hw_ble/ stack/ # scanner, gatt, sm codec/ # sfloat, float, crc8, ticd_checksum normalize/ # reading, quality
session/ # episodic, stream orchestrators drivers/ beurer_bm54/ aand_ua651/ masimo_mightysat/ ticd_thermometer/
htp_thermometer/ nipro_cocoron/ fora_connect/ # blocked stub hub/ # patient bind, mqtt
```

10.6 What NOT to reuse naively

Anti-reuse	Reason
Single "BP driver" for Beurer+A&D without arming strategy	Will miss A&D dumps
Single "thermometer driver" for TICD+HTP	Different transports and frames
Service UUID-only discovery	Breaks MightySat variants
Global keep-alive ping	Not in corpus; may break episodic devices
Auto memory clear after any receive	Risk of data loss on northbound failure
Assuming Just Works everywhere	Beurer passkey; A&D encryption; classic PINs

11. Red Flags, Anomalies & Implementation Risks

ID	Severity	Item	Mitigation
R1	Critical	FORA has no protocol in corpus	Block driver; obtain SDK/sniff
R2	Critical	A&D 5-second CCCD+time gate	Arm immediately after encrypt; precompute writes
R3	Critical	Beurer User ID rev mismatch	Version-detect; map correctly
R4	High	Beurer Passkey required on newer units	Implement Passkey Entry UX/automation
R5	High	MightySat service UUID optional in adv	Filter CID 0x0243
R6	High	TICD dual-wake + unsolicited 0x54	State machine with ignore rules
R7	High	Cocoron always-on vs BP episodic	Separate connection managers
R8	High	Cocoron on-wire UUID/payload incomplete	Need IF workbook or sniffer
R9	Medium	Beurer sends both users always	PatientRouter must filter
R10	Medium	A&D rejects settings during dump	Don't interleave commands mid-indicate
R11	Medium	MightySat PI not in trends	Don't expect historical PI
R12	Medium	Invalid trend value 255	Explicit INVALID quality
R13	Medium	TD1241 command subset	Model capability matrix
R14	Medium	No battery on Beurer protocol	UI must not assume Battery Service
R15	Low	Beurer MAP always 0	Don't validate MAP clinically
R16	Low	HTP idle 5 s	Finish config quickly
R17	High	Official apps may do undocumented extras	Validate with HCI sniff before production

12. Documentation Gaps & Capture Plan

Gap	Impact	Recommended action
FORA GATT/commands/pairing	Cannot integrate	Request ForaCare protocol; sniff iFORA
Cocoron UUID & Huffman tables	ECG parse blocked	Obtain interface design xlsx; sniff app
MightySat LE security level	Pairing uncertainty	Sniff official app pairing features
Beurer exact CCCD timing / disconnect initiator	Minor session polish	Sniff HealthManager-style app
A&D full command binary examples beyond partial table	Custom service completeness	Lab exercise all cmd IDs
Whether any device needs explicit L2CAP conn_param_update	Throughput/stability	Log official app LL/L2CAP
Multi-phone policies except Cocoron warning	Fleet deployments	Test second central behavior

Appendix A — UUID & Constant Catalog

A.1 SIG UUIDs used

Name	UUID
GAP	0x1800
GATT	0x1801
Device Information	0x180A
Battery	0x180F
Blood Pressure	0x1810
Pulse Oximeter (PLXP)	0x1822
Device Name	0x2A00
Appearance	0x2A01
Peripheral Preferred Connection Parameters	0x2A04
Service Changed	0x2A05
Date Time	0x2A08
Battery Level	0x2A19
System ID	0x2A23
Model Number	0x2A24
Serial Number	0x2A25
Firmware Revision	0x2A26
Hardware Revision	0x2A27
Software Revision	0x2A28
Manufacturer Name	0x2A29
IEEE Regulatory Cert List	0x2A2A
Blood Pressure Measurement	0x2A35
Intermediate Cuff Pressure	0x2A36
Blood Pressure Feature	0x2A49
PnP ID	0x2A50

CCCD	0x2902
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A.2 Proprietary UUIDs / IDs

Brand	Constant	Value
Beurer	Company ID	0x0611
Beurer	Adv name	BM54
A&D	Custom service	0xF000
A&D	Custom char family	233BF001-5A34-1B6D-975C-000D5690ABE4 (as mapped)
Masimo	Company ID	0x0243
Masimo	Service	54c21000-a720-4b4f-11e4-9fe20002a5d5
Masimo	RX / TX	0x1001 / 0x1002
Masimo	SOM	0x77
TICD BLE	UUID base	1212-efde-1523-785feabcd123
TICD BLE	Service / Char	0x1523 / 0x1524
TICD	Frame start	0x51
TICD	Stop GW→MD / MD→GW	0xA3 / 0xA5

A.3 CCCD values

Value	Meaning
0x0000	Disable
0x0001	Notifications
0x0002	Indications

Appendix B — Command Catalogs

B.1 MightySat

ID	Name
0x01	Get Device Information
0x02	Set Clock
0x03	Configure Streaming Data
0x04	Waveform Streaming Response (device)
0x05	Parameter Streaming Response (device)
0x06	Get Trend Record
0x07	Clear Trend Records
0xFE	ACK
0xFF	NACK

B.2 TICD thermometer

ID	Name
0x23	Read clock
0x24	Read model
0x25	Read storage time
0x26	Read storage result
0x27	Read SN part1
0x28	Read SN part2
0x2B	Read storage count
0x33	Write clock
0x41	Start IR measure
0x50	Power off
0x52	Clear memory
0x54	Enter comm mode notify

B.3 A&D custom (UA-651BLE)

ID	Name
0x01	Set Time
0x03	Disconnect / stand-by request
0x10	Unpair
0x12	Delete all memory
0xA6	Memory/buffer number related
0xD6	Read buffer size
0xE1	Request all memory data (sequence diagrams)

Appendix C — Source Document Index

Path	Brand	Used for
beurer/BM54_transmissionprotocol_rev03-20240716.pdf	Beurer	Primary BM54 protocol
beurer/BM54_transmissionprotocol_rev01-20190805.pdf	Beurer	User ID delta / history
beurer/BM54_transmissionprotocol_rev02-20200430.pdf	Beurer	Intermediate revision
FORA/FORA 6 Connect Brochure-Ver1.0_20250407.pdf	FORA	Product capabilities only
nipro/SH3/■■■■■■■■sdk_ble_ua-651ble_V1.2_141027_-En.pdf	A&D	Full BP BLE SDK
nipro/SH3/■SpO2■MightySat ... CSD-1322B.pdf	Masimo	Full SpO2 protocol
nipro/SH3/■■■■■■■■1-1■■... TICD_Thermometer_v1.16...pdf	Thermometer	Serial/BLE commands
nipro/SH3/■■■■■■■■1-2■■... HTP_V10.pdf	SIG	HTP collector rules
nipro/SH3/Cocoron■■■■■■.pdf	Nipro ECG	Always-on ECG design

nipro/SH3/SH3_Features.pdf	SH3 hub	Platform/MQTT context
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Document Control

Item	Value
Title	BLE Medical Device Hardware-Layer Architecture Reference
Version	1.0
Based on local datasheets only	Yes
Code included	No
Completeness intent	All extractable protocol facts from corpus + architecture synthesis

End of document.